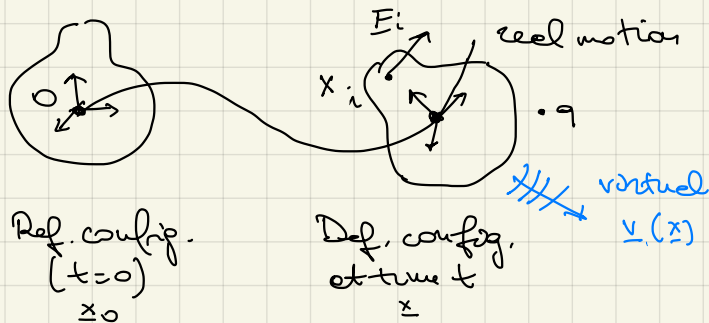

lectures 13, 14

PPV = Theorem of virtual power

19, 21 / x1 / 25 

Principle of virtual $\int$ power (PPV)

Recall the version for rigid bodies



$$\underline{\mathcal{F}} = \sum \underline{F}_i = 0$$

$$\underline{M}_q = \sum (x_i - q) \wedge \underline{F}_i = 0$$

loads $\underline{F}_i, i=1, \dots, m$

$$\left. \begin{aligned} \underline{\mathcal{F}} = \sum \underline{F}_i = 0 \\ \underline{M}_q = \sum (x_i - q) \wedge \underline{F}_i = 0 \end{aligned} \right\} \Leftrightarrow P_{\text{ext}}[L](\underline{v}) = 0$$

$\forall \underline{v}$ virtual rigid velocity field
= admissible test field

= virtual velocity field on "test" velocity field

$\underline{v}(x)$ = field of velocities in a virtual rigid motion away from current / deformed configuration.

$$= \underline{v}_q + \underline{\omega} \wedge (x - q) \quad \text{in which } \underline{v}_q \text{ and } \underline{\omega} \text{ are arbitrary}$$

i.e., let forces expend power against it

$$\sum \underline{F}_i \cdot \underline{v}(x_i) = \sum \underline{F}_i \cdot (\underline{v}_q + \underline{\omega} \wedge (x_i - q)) = \underbrace{\underline{v}_q \cdot \sum \underline{F}_i}_{\underline{\mathcal{F}}} + \underbrace{\underline{\omega} \cdot \sum (x_i - q) \wedge \underline{F}_i}_{\underline{M}_q} = 0 \quad \forall \underline{v}_q, \underline{\omega} \Leftrightarrow \begin{aligned} \underline{\mathcal{F}} &= 0 \\ \underline{M}_q &= 0 \end{aligned}$$

[in particular, in 2D,
 $\underline{\omega} = \omega \underline{e}_3, \underline{M}_q = M_q \underline{e}_3$

Idea: instead of saying that something = 0, one says that it's orthogonal to every test vector, i.e., its scalar product with every test vector is = 0.

test vector = virtual velocity

scalar product = power expended

Remark: PPV provides a test to check whether a loaded mechanism can be in equilibrium under the given loads.

$$\underline{x} \text{ equilibrium config. under loads } L \Leftrightarrow P_{\text{ext}}[L](\underline{\xi}^*) = 0 \quad \forall \underline{\xi}^* \text{ kinematically admissible velocity}$$

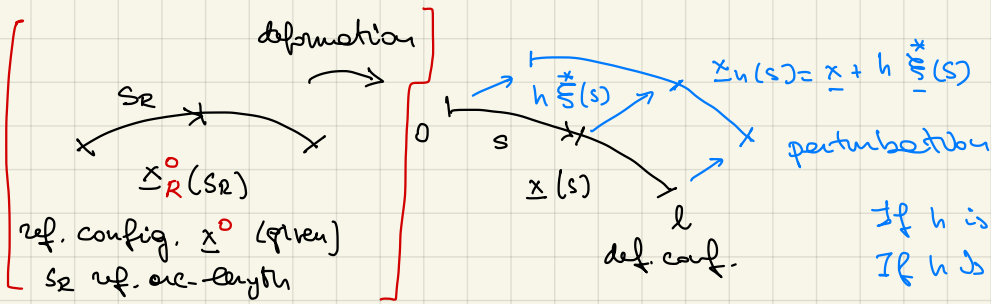
(perturbation $\underline{x}_h = \underline{x} + h \underline{\xi}^*, h \ll 1$)

One can generalize this idea to force densities and virtual velocity fields
we'll get in this way a variational formulation of the equilibrium equations,
based on computing the power expended against virtual variations of
the current configuration.

Remark: kinematically admissible virtual velocity field = compatible with the constraints
Reactions of frictionless constraints do not contribute to power
Test for equilibrium for mechanisms under ext. loads that expend no power for every kin. admissible velocity field.

Executive summary for PPV/PLV

Kinematics of virtual displacement fields for planar rods



If h is time, $\underline{\xi}^*$ is velocity (virtual)
If h is non-dim., $\underline{\xi}^*$ is displacement (virtual)

Any quantity G [say, gravitational energy] depending on $\underline{x}$ becomes a function G_h of h through $\underline{x}_h$

$$G_h = G(\underline{x}_h) = G(\underline{x} + h^* \underline{\xi})$$

Consider $\frac{d}{dh} G_h \Big|_{h=0} =: \dot{G} = \delta G$ (It's the "directional derivative" along $\underline{\xi}^*$)
(variational, calculus of variations, Euler)

$$\text{e.g. } G_h = G(\underline{x}_h) = \int_0^l \rho(s) g \underline{x}_{h2}(s) ds = \int_0^l \rho g \underline{x}_2 ds + h^* \int_0^l \rho g \underline{\xi}_2 ds = \dot{G}$$

Quantities of interest: $\underline{t}_h, \underline{n}_h, \alpha_h, s_h$ from $\underline{s}'_h = |\underline{x}'_h|$, $c_h = \frac{\alpha'_h}{s'_h}$ (geometry)
 $\underline{\xi}_h, u_h = \underline{\xi}_h \cdot \underline{t}, v_h = \underline{\xi}_h \cdot \underline{n}$
 $\omega_h = \alpha_h - \alpha, \epsilon_h = s'_h - 1, k_h = \omega'_h$ } deformation from $\underline{x}$ to $\underline{x}_h$

For each of these quantities can compute $\frac{d}{dh} G_h = \dot{G}$. After some calculations

$$\left[\dot{\underline{s}} = \dot{u} \underline{t} + \dot{w} \underline{n}, \quad \dot{\underline{s}}' = \dot{\epsilon} \underline{t} + \dot{\omega} \underline{n}, \quad \dot{k} = \dot{\omega}' \right]$$

key equations
(kinematic compatibility linking virtual strains $\dot{\epsilon}, \dot{k}$ with virtual displacements $\dot{u}, \dot{w}, \dot{\omega}$)

$$\left[|\underline{s}'|^* = (1 + \epsilon)^* = \epsilon^* \right] \quad \text{skip}$$

Then are reminiscent of linear kinematics for straight rods $\underline{t}' = 0, \underline{n}' = 0$

$$\dot{\underline{s}} = \dot{u} \underline{t} + \dot{w} \underline{n}, \quad \dot{\underline{s}}' = \dot{u}' \underline{t} + \dot{w}' \underline{n} = \dot{\epsilon} \underline{t} + \dot{\omega} \underline{n}, \quad \dot{k} = \dot{\omega}'$$

$$\underline{\epsilon} = \dot{u}', \quad \underline{\omega} = \dot{w}', \quad \underline{k} = \dot{\omega}''$$

[For derivation of these formulas see: Complements]

From virtual displacements/velocities to virtual work/power

Given $s \mapsto \underline{\xi}^*(s)$ field (virtual displacement/velocities; test field)

$$\underline{\xi}^* = \underline{\xi} \underline{t} + \underline{\omega} \underline{\eta}, \text{ hence } \underline{\omega}(s) = \underline{\xi}^{*'} \cdot \underline{\eta}, \underline{\xi}(s) = \underline{\xi}^* \cdot \underline{t}, \underline{\dot{\xi}}^* = \underline{\dot{\omega}}^* \quad [\text{kin. comp.}]$$

Do the case without discontinuities of $\underline{\xi}^*$, or of M, N, T first

Assume $\underline{x}(s)$ is an equilibrium config. under

loads $\underline{F}_0, M_0, \underline{f}, m, \underline{F}_l, M_l$ (L)

with internal "forces" M, N, T

Equilibrium: $0 = \underline{R}' + \underline{f}$
 $0 = M' + T + m$

BC's $\underline{R}(0) = -\underline{F}_0 \quad \underline{R}(l) = \underline{F}_l$
 $M(0) = -M_0 \quad M(l) = M_l$

$$0 = \int_0^l (\underline{R}' \cdot \underline{\xi}^* + \underline{f} \cdot \underline{\xi}^* + M' \underline{\omega}^* + T \underline{\omega}^* + m \underline{\omega}^*) ds = (*)$$

$$\int_0^l \underline{R}' \cdot \underline{\xi}^* ds = \int_0^l [(\underline{R} \cdot \underline{\xi}^*)' - \underline{R} \cdot \underline{\xi}^{*'}] ds = \underline{R}(l) \cdot \underline{\xi}^*(l) - \underline{R}(0) \cdot \underline{\xi}^*(0) - \int_0^l \underline{R} \cdot \underline{\xi}^{*'} ds$$

$$= \underline{F}_l \cdot \underline{\xi}^*(l) + \underline{F}_0 \cdot \underline{\xi}^*(0) - \int_0^l (N \underline{\xi}^* + T \underline{\omega}^*) ds$$

$$\int_0^l M' \underline{\omega}^* ds = \int_0^l [(M \underline{\omega}^*)' - M \underline{\omega}^{*'}] ds = \underline{M}(l) \underline{\omega}^*(l) - \underline{M}(0) \underline{\omega}^*(0) - \int_0^l M \underline{\dot{\xi}}^* ds \quad [\underline{\omega}^* = \underline{\dot{\xi}}^*]$$

$$(*) = \underline{F}_l \cdot \underline{\xi}^*(l) + M_l \underline{\omega}^*(l) + \underline{F}_0 \cdot \underline{\xi}^*(0) + M_0 \underline{\omega}^*(0) \quad (a)$$

$$+ \int_0^l (\underline{f} \cdot \underline{\xi}^* + m \underline{\omega}^*) ds$$

$$- \int_0^l (N \underline{\xi}^* + M \underline{\dot{\xi}}^*) ds \quad (b)$$

$$= 0 + \underline{\xi}^* \text{ virtual } \left\{ \begin{array}{l} \text{displacement} \\ \text{velocity} \end{array} \right\} \text{ field} = \text{test field}$$

$$(a) = P_{\text{ext}} [\underline{F}_l, M_l, \underline{F}_0, M_0, \underline{f}, m] (\underline{\xi}^*)$$

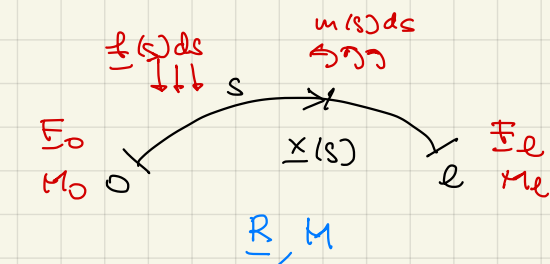
L loads

external work/power

$$(b) = P_{\text{def}} [M, N] (\underline{\xi}^*)$$

internal or deformation work/power

(it would be = 0 if $\underline{\xi}^*$ wgsd)



$$\left. \begin{array}{l} \underline{\xi}^*(s) \\ \underline{\omega}^*(s) \end{array} \right\} \text{ sum and integrate } \int_0^l ds$$

"test" equl. eqn's by multiplying by test functions (virtual work/power)

$$N \underline{t} + T \underline{\eta}$$

$$\underline{\xi} \underline{t} + \underline{\omega} \underline{\eta}$$

and use BC's

$$+ \int_0^l T \underline{\omega}^* ds - \int_0^l T \underline{\omega}^* ds$$

PPV (a theorem!)

$\underline{x}(s)$ equilibrium configuration under loads $\underline{L}$
with internal "forces" M, N, T
(shears)

$$\Rightarrow P_{\text{ext}}[\underline{L}](\underline{\xi}^*) = P_{\text{def}}[M, N](\underline{\xi}^*) \quad \forall \underline{\xi}^*$$

arbitrary test field / perturbation

In fact, also the opposite implication $\Leftarrow$ is true.

This requires to know that $\int_0^l u(s) \varphi(s) ds = 0 \quad \forall \varphi \Rightarrow u(s) \equiv 0$
which φ 's? Fundamental lemma of Calculus of Variations

Using this fact, by following the argument (integration by parts) in the opposite direction we get

$$0 = \int_0^l (\underline{R}' \cdot \underline{\xi}^* + \underline{f} \cdot \underline{\xi}^* + M' \tilde{\omega}^* + T \tilde{\omega}^* + m \tilde{\omega}^*) ds \quad \forall \underline{\xi}^* \Rightarrow \begin{aligned} \underline{R}' + \underline{f} &= 0 \\ M' + T + m &= 0 \end{aligned}$$

The integral identity $P_{\text{ext}}[\underline{L}](\underline{\xi}^*) = P_{\text{def}}[M, N](\underline{\xi}^*) \quad \forall \underline{\xi}^*$ admissible test field
is at the basis of the most wide-spread numerical procedure to solve the
differential equations of equilibrium: weak form of differential equations,
GALERKIN method (projection onto a finite-dimensional function space),
Finite Element Method (FEM).

(*) with some subtleties due to the fact that $\tilde{\omega}^* = \underline{\xi}^* \cdot \underline{n}$ is NOT indep. of $\underline{\xi}^*$,
see Merigo.

Recap on PPV

- PPV for rigid systems

$$\underline{\dot{g}}(L) = \underline{0}$$

$$\underline{m}_0(L) = \underline{0}$$

$$\Leftrightarrow P_{\text{ext}}[L](\underline{\dot{\xi}}^*) = 0 \quad \forall \underline{\dot{\xi}}^* \text{ test}$$

(virtual rigid velocity)

- PPV for deformable systems

$$\underline{R}' + \underline{f} = \underline{0}$$

$$\underline{M}' + \underline{T} + \underline{m} = \underline{0}$$

$\Leftrightarrow$

$$P_{\text{ext}}[L](\underline{\dot{\xi}}^*) = P_{\text{int}}[M, N](\underline{\dot{\xi}}^*) \quad \forall \underline{\dot{\xi}}^*$$

(internal forces balance)
external loads

(external virtual power balances)
internal/def. virtual power

$$P_{\text{int}}[M, N](\underline{\dot{\xi}}^*) = \int_0^L (\underline{M} \underline{\dot{\kappa}}^* + \underline{N} \underline{\dot{\epsilon}}^*) ds = \int_0^{L_R} (\underline{M}_R \underline{\dot{\kappa}}_R^* + \underline{N}_R \underline{\dot{\epsilon}}_R^*) (1 + \epsilon_R) dS_R$$

- Deformation/internal power density

$$(\underline{M} \underline{\dot{\kappa}}^* + \underline{N} \underline{\dot{\epsilon}}^*) ds$$

per unit current length

$$(\underline{M}_R \underline{\dot{\kappa}}_R^* + \underline{N}_R \underline{\dot{\epsilon}}_R^*) (1 + \epsilon_R) dS_R$$

per unit reference length

$$\underline{\dot{\epsilon}}_R (1 + \epsilon_R) = \underline{\dot{\epsilon}}_R^0$$

The term $\underline{\dot{\epsilon}}_R (1 + \epsilon_R)$ represents, in fact, the virtual rate of increment of extensional strain, measured with respect to the reference configuration, associated with the perturbation $\underline{x} \rightarrow \underline{x}_h$.

$$\text{Similar interpretation for } \underline{\dot{\kappa}}_R (1 + \epsilon_R) = \underline{\dot{\kappa}}_R^0$$

- with conservative loads and hyperelastic constitutive equations

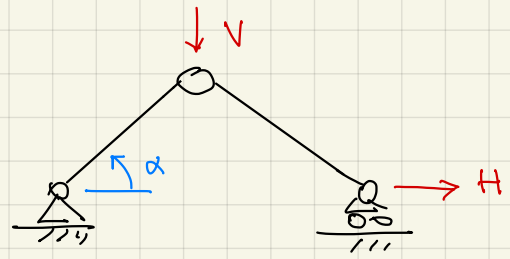
PPV becomes principle of minimal/stationary potential energy.

Generalizations

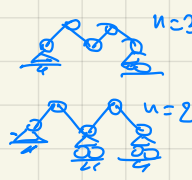
1) PPV/PLV for rigid bodies

$\underline{x}$ of equilibrium under loads $\mathcal{L} \iff P_{ext}[\mathcal{L}](\underline{\dot{x}}^*) = 0 \neq \underline{\dot{x}}^*$ kin. adm. velocity in perturbation from $\underline{x}$
 $(\underline{x}_h = \underline{x} + h \underline{\dot{x}}^*)$

2) PPV/PLV for piecewise rigid, articulated systems subject to non-dissipative constraints (ext. and int.) (and conservative forces)



Effective DoFs: $\underline{q} = (q_1, \dots, q_n)$ n DoFs
 eg. α $n=1$



$\underline{q}$ of equilibrium under loads $\mathcal{L} \iff P_{ext}[\mathcal{L}](\underline{\dot{q}}^*) = 0 \neq \underline{\dot{q}}^*$ kin. adm. velocity in perturbation from $\underline{q}$

[Lagrange late '700, Courcy, Desautel Venant, Müller Breslau, Mohr '800, FEH '900]
 descriptive variables: late '700 and 1980's
 $(\underline{q}_h = \underline{q} + h \underline{\dot{q}}^*)$

Conservative forces: positional forces with potential energy V_{ext} s.t.

$$P_{ext}[\mathcal{L}](\underline{\dot{q}}^*) = -\nabla_{\underline{q}} V_{ext}(\underline{q}) \cdot \underline{\dot{q}}^*$$

$\underline{q}$ equilibrium under loads $\mathcal{L} \iff P_{ext}[\mathcal{L}](\underline{\dot{q}}^*) = -\nabla_{\underline{q}} V_{ext}(\underline{q}) \cdot \underline{\dot{q}}^* = 0 \neq \underline{\dot{q}}^* \Rightarrow \nabla_{\underline{q}} V_{ext}(\underline{q}) = 0$

$\underline{q}$ is ^{stable} _{unstable equilibrium} if V_{ext} has a local ^{minimum} _{maximum} at $\underline{q}$. (Principle of minimal potential energy)

Special case: dead loads (forces indep. of shape of the point of application)

$$V_{ext}(\underline{q}) = -W_{ext}[\mathcal{L}](\underline{q}_0 \rightarrow \underline{q})$$

work done by the loads when going from $\underline{q}_0$ to $\underline{q}$

3) PPV/PLV for deformable systems

• $\underline{x}$ of equilibrium under loads $\underline{L}$ and internal forces $\underline{L}_{int}$ $\leftrightarrow \mathcal{P}_{ext}[\underline{L}](\underline{\xi}^*) = \mathcal{P}_{def}[\underline{L}_{int}](\underline{\xi}^*) \quad \forall \underline{\xi}^*$

For 2d rods, (linear const. equations) this is

$$\mathcal{P}_{ext}[\underline{f}, \underline{F}_0, \underline{F}_L](\underline{\xi}^*) = \int_0^{l_R} \underline{f}_R(\sigma) \cdot \underline{\xi}_R^*(\sigma) d\sigma + \underline{F}_0 \cdot \underline{\xi}_R^*(0) + \underline{F}_L \cdot \underline{\xi}_R^*(l_R) =$$

$$\mathcal{P}_{def}[\underline{M}, \underline{N}](\underline{\xi}^*) = \int_0^{l_R} (M_R(\sigma) \underline{\kappa}_R^*(\sigma) + N_R(\sigma) \underline{\varepsilon}_R^*(\sigma))(1 + \varepsilon_R(\sigma)) d\sigma$$

where $\underline{\varepsilon}_R^* = \underline{\xi}_R^{*'} \cdot \underline{t}$, $\underline{\kappa}_R^* = \underline{\omega}_R^{*'}$

; $\underline{\varepsilon}_R^* = \underline{u}'$, $\underline{\kappa}_R^* = \underline{w}''$ in linear kinematics

kin. adm. velocity
in perturbed state

$$(\underline{x}_u = \underline{x} + h \underline{\xi}^*)$$

When internal forces are elastic and conservative, and the system has finitely many DOFs

$$\mathcal{P}_{def}[\underline{L}_{int}](\underline{q}^*) = \nabla_{\underline{q}} V_{def}(\underline{q}) \cdot \underline{q}^*$$

and we get $0 = \mathcal{P}_{ext}[\underline{L}](\underline{q}^*) - \mathcal{P}_{def}[\underline{L}_{int}](\underline{q}^*) = \underbrace{(-\nabla V_{ext}[\underline{L}] - \nabla V_{def}[\underline{L}_{int}](\underline{q})) \cdot \underline{q}^*}_{=0}, \quad \forall \underline{q}^*$

or, setting $V(\underline{q}) = V_{ext}(\underline{q}) + V_{def}(\underline{q})$,

.. $\underline{q}$ of equil. under loads $\underline{L}$ and internal forces $\underline{L}_{int}$ $\leftrightarrow \nabla_{\underline{q}} V(\underline{q}) = 0$

... $\underline{q}$ is ^{stable equilibrium} if V has a local ^{minimum} at $\underline{q}$. (Principle of minimal potential energy)
unstable equilibrium ^{maximum}

PPV as an "algorithm": $\begin{cases} \text{compute elastic displacements in small deformation theory (\# 1)} \\ \text{establish principle of minimal/stationary potential energy (\# 2)} \\ \text{Finite Element Method (FEM) (\# 3)} \end{cases}$

Recall PPV

S_L statically admissible system or $\underline{x}(s)$: L, M, N, T satisfy

- differential equations of equilibrium (e.g., $E' + f = 0$ etc.)
- static BCs

S^* kinematically admissible system of perturbations for $\underline{x}(s)$: $\underline{\xi}^*, \underline{\tilde{\omega}}^*, \underline{\tilde{\epsilon}}^*, \underline{\tilde{\kappa}}^*$ satisfy

- differential equations of kinematics (e.g., $\underline{\tilde{\kappa}}^* = \underline{\tilde{\omega}}'^*$ etc.)
- kinematic BCs

Then $P_{ext}[L](\underline{\xi}^*) = P_{int}[M, N](\underline{\xi}^*) + \underline{\xi}^*$ kinematically admissible (*)

Using constitutive equations

$$M = EI \kappa$$

$$N = EA \epsilon$$

and setting

$$\underline{\tilde{M}}^* = EI \underline{\tilde{\kappa}}^*$$

$$\underline{\tilde{N}}^* = EA \underline{\tilde{\epsilon}}^*$$

can write

from S_L stat. admis.

$$\begin{aligned} & \int_0^l \left(\underline{f} \cdot \underline{\xi}^* + m \underline{\tilde{\omega}}^* \right) ds \\ & + \sum_{s_i} \underline{F}_{s_i} \cdot \underline{\xi}^*(s_i) + M_{s_i} \underline{\tilde{\omega}}^*(s_i) \\ & \left[= \int_0^l (M \underline{\tilde{\kappa}}^* + N \underline{\tilde{\epsilon}}^*) ds = \right] \\ & = \int_0^l \left(\frac{M M^*}{EI} + \frac{N N^*}{EA} \right) ds \end{aligned}$$

from S^* kin. admis.

This is necessary and sufficient condition (N&S)
for $\underline{x}(s)$ is an equilibrium conf., under S_L

Special case: linearized theory

Small perturbations from straight ref. config.

$$\underline{x}(s) = 0 + s \underline{e}_1$$

$$M = EI \kappa = EI w''$$

$$N = EA \epsilon = EA u'$$

$$\underline{\tilde{\xi}}^* = \underline{\tilde{u}}^* \underline{e}_1 + \underline{\tilde{w}}^* \underline{e}_2$$

$$\underline{\tilde{\omega}}^* = \underline{\tilde{w}}'^*$$

$$\underline{\tilde{\kappa}}^* = \underline{\tilde{\omega}}''^* = \underline{\tilde{M}}^*/EI$$

$$\underline{\tilde{\epsilon}}^* = \underline{\tilde{u}}'^* = \underline{\tilde{N}}^*/EA$$

N&S for $\underline{x}(s) = 0 + s \underline{e}_1$

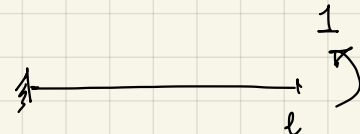
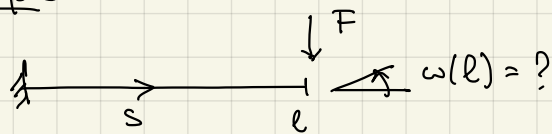
equilibrium configuration under S_L

Use as a tool to evaluate (small def.) elastic displacements produced by loads on straight beam in equil. config.

interpret these as a known, admissible perturbation of equil. conf. $\underline{x}(s) = \underline{0} + s\mathbf{e}_1$

$$\underline{x}(s) = \underline{0} + s\mathbf{e}_1$$

Example



$$\mathbf{S} = \mathbf{S}^{(1)} = \mathbf{S}^*$$

$$\text{Consider } \mathbf{S}^{(2)} = \mathbf{S}_\ell$$

Since $\underline{x}(s) = \underline{0} + s\mathbf{e}_1$ is an equil. configuration under loads of $\mathbf{S}_\ell$

and small elastic displacements in $\mathbf{S} = \mathbf{S}^*$ are kinematically admissible, we have

$$P_{\text{ext}}[L](\underline{\xi}^*) = P_{\text{def}}[M, N](\underline{\xi}^*)$$

from $\mathbf{S}_\ell = \mathbf{S}^{(2)}$

from $\mathbf{S}^* = \mathbf{S}^{(1)}$

$$1 \cdot \omega(l) = \int_0^l \left(\frac{M M^*}{EI} + \frac{N N^*}{EA} \right) ds = \int_0^l \left(\frac{M^{(2)} M^{(1)}}{EI} + \frac{N^{(2)} N^{(1)}}{EA} \right) ds$$

If we can determine $M^{(1)}, M^{(2)}, N^{(1)}, N^{(2)}$ (and we can, see later), then we can calculate $\omega(l)$ by evaluating the integrals on RHS.

Idea: let $\underline{\xi}, \delta = \int_{\omega}^u$ be the (small def.) elastic displacements produced by loads (in $\mathbf{S}^*$)

in equilibrium on straight beam $\underline{x}(s) = \underline{0} + s\mathbf{e}_1$. Interpret these unknown displacements

$\underline{\xi}, \delta$ as kinematically admissible perturbations of $\underline{x}(s)$ that can be used to test any $\mathbf{L}$

for equilibrium. Use $\mathbf{L}$ such that

$$P_{\text{ext}}[L](\underline{\xi}^*) = 1 \delta$$

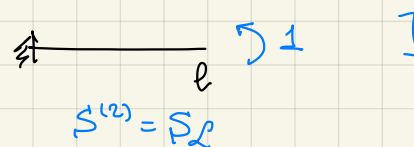
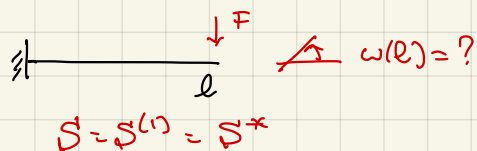


i.e., the loads in $\mathbf{S}_\ell$ must "test" for the required unknown displacement

This leads to a first algorithm, #1.

Algorithm #1

[Example:



1. Choice of S^*

Consider unknown displacements ξ, δ as a kinem. admiss. particular for ref. conf. $\underline{x}(s) = \underline{0} + s$

$$\Rightarrow S = S^{(1)} \equiv S^*, \text{ hence } \xi = \xi^*, \delta = \delta^*$$

2. Choice of S_L

Interpret $\underline{x}(s) = \underline{0} + s \underline{e}_1$ as equil. conf. p. for a system of loads S_L with the property that

$$P_{ext}[L](\xi^*) = 1 \delta$$

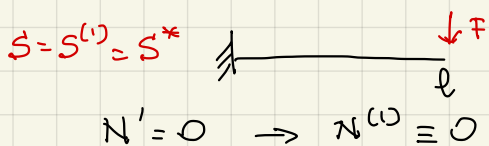
$$\Rightarrow S_L = S^{(2)} \text{ stat. admiss. "test" system, i.e., s.t. } P_{ext}[L](\xi^*) = 1^{(2)} \cdot \delta^{(1)}$$

3. write PIV using $S^{(2)}$ as stat. admiss. and $S^{(1)}$ as kin. admiss. systems

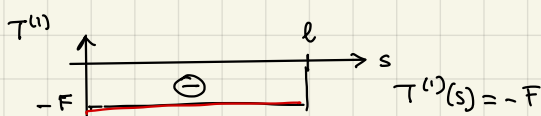
$$1 \cdot \delta = 1^{(2)} \cdot \delta^{(1)} = P_{ext}[L](\xi^*) = P_{def}[M, N](\xi^*) = \int_0^l \left(M^{(2)} \frac{M^{(1)}}{EI} + N^{(2)} \frac{N^{(1)}}{EA} \right) ds$$

4. Read off δ from the identity in 3, after having calculated $M^{(2)}, M^{(1)}, N^{(2)}, N^{(1)}$

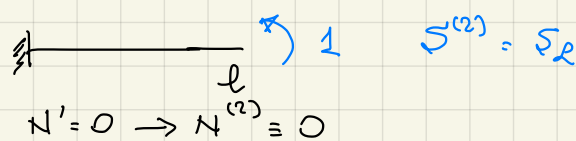
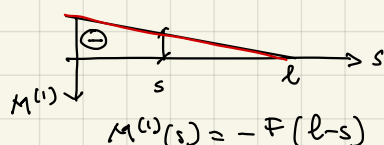
In the example above we have



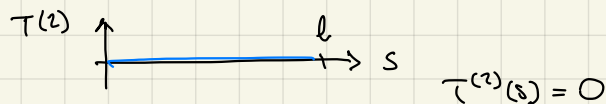
$$T' = 0 \rightarrow T(s) = \text{const} = T(l) = -F$$



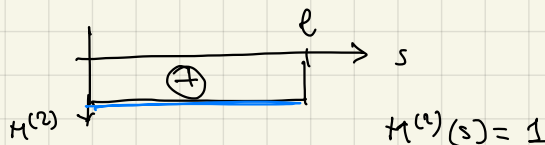
$$M' = -T = F \rightarrow M(s) \text{ linear, } M(l) = 0$$



$$T' = 0 \rightarrow T(s) = \text{const} = T(l) = 0$$



$$M' = -T = 0 \Rightarrow M(s) = \text{const} = M(l) = 1$$



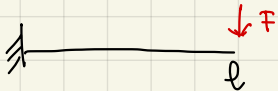
(convention on diagram of M : riportare ordinata del lato delle fibre tese)

positive bending moment stretches bottom fibers

← Need some comments

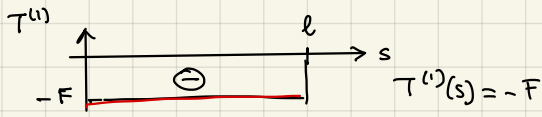
Example continued

$S = S^{(1)} = S^*$

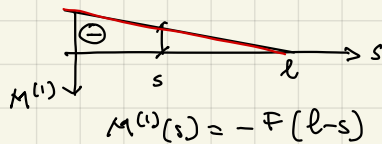


$N' = 0 \rightarrow N^{(1)} \equiv 0$

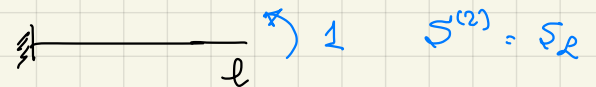
$T' = 0 \rightarrow T(s) = \text{const} = T(l) = -F$



$M' = -T = F \rightarrow M(s) \text{ linear, } M(l) = 0$

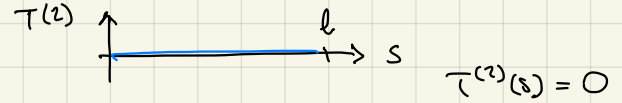


$S^{(2)} = S_l$

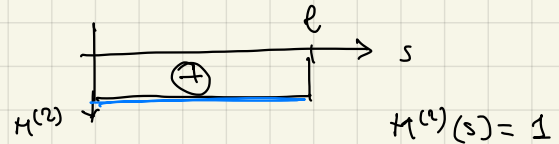


$N' = 0 \rightarrow N^{(2)} \equiv 0$

$T' = 0 \rightarrow T(s) = \text{const} = T(l) = 0$



$M' = -T = 0 \Rightarrow M(s) = \text{const} = M(l) = 1$



(convention on diagram of M : riportare ordinate del lato delle fibre tese)
 positive bending moment stretches bottom fibres

← Need some comments

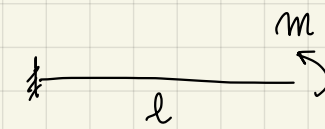
$$\frac{1}{EI} \omega(l) = \int_0^l M^{(2)} \frac{M^{(1)}}{EI} ds = \int_0^l 1 \frac{-F(l-s)}{EI} ds = -\frac{F}{EI} \left(l^2 - \frac{s^2}{2} \right) \Big|_0^l = -\frac{F}{EI} \left(l^2 - \frac{l^2}{2} \right) = -\frac{Fl^2}{2EI}$$

i.e. $\omega(l) = -\frac{Fl^2}{2EI}$. Sign OK (negative means clockwise)

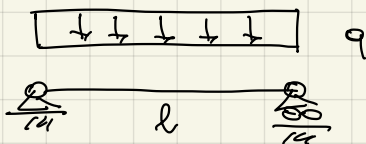
Dimensions OK $\left(\frac{FL^2}{FL^{-2}L^4} = \text{radians} \right)$

← Need some comments

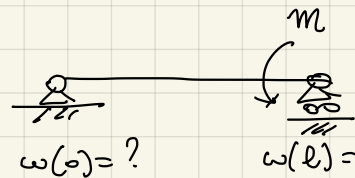
Examples of exercises (we'll do them in class)



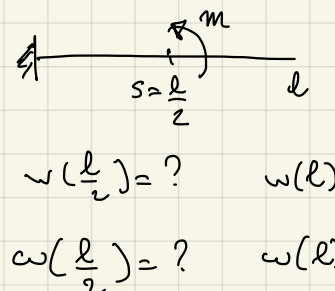
$w(l) = ?$
 $\omega(l) = ?$



$w(0) = ?$ $w(l) = ?$
 $w(\frac{l}{2}) = ?$



$w(0) = ?$ $w(l) = ?$



$w(\frac{l}{2}) = ?$ $w(l) = ?$
 $\omega(\frac{l}{2}) = ?$ $\omega(l) = ?$

Algorithm #2

look at FPI - Principle of minimal potential energy reinterpretation
of case of rigid beam under centrifugal force (Complements)